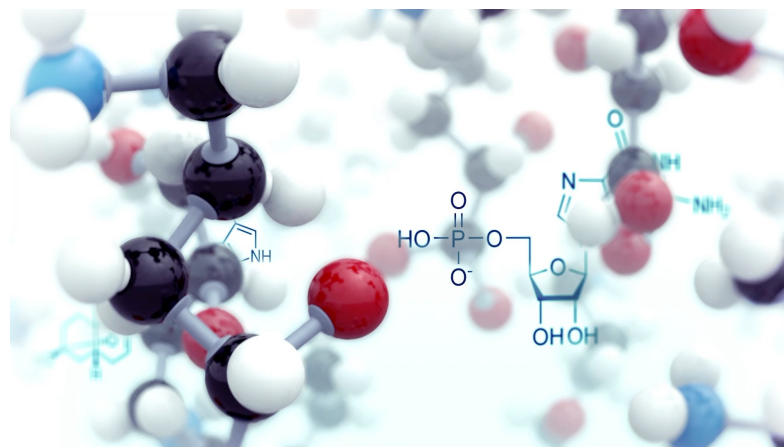


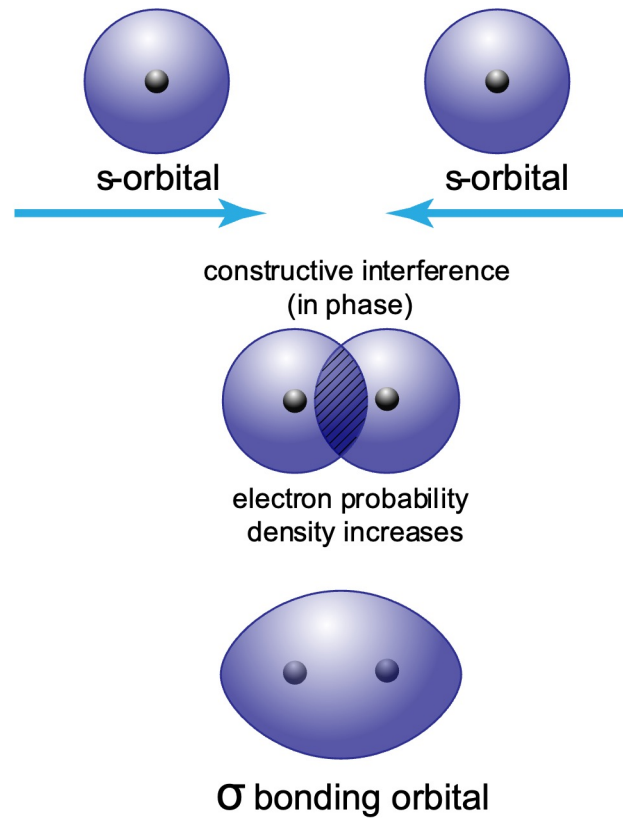
Molecular orbitals of Oxygen, Correlation of pKa of acids with physical parameters, free energy of a reaction using Computational Modelling

Experiment # 3

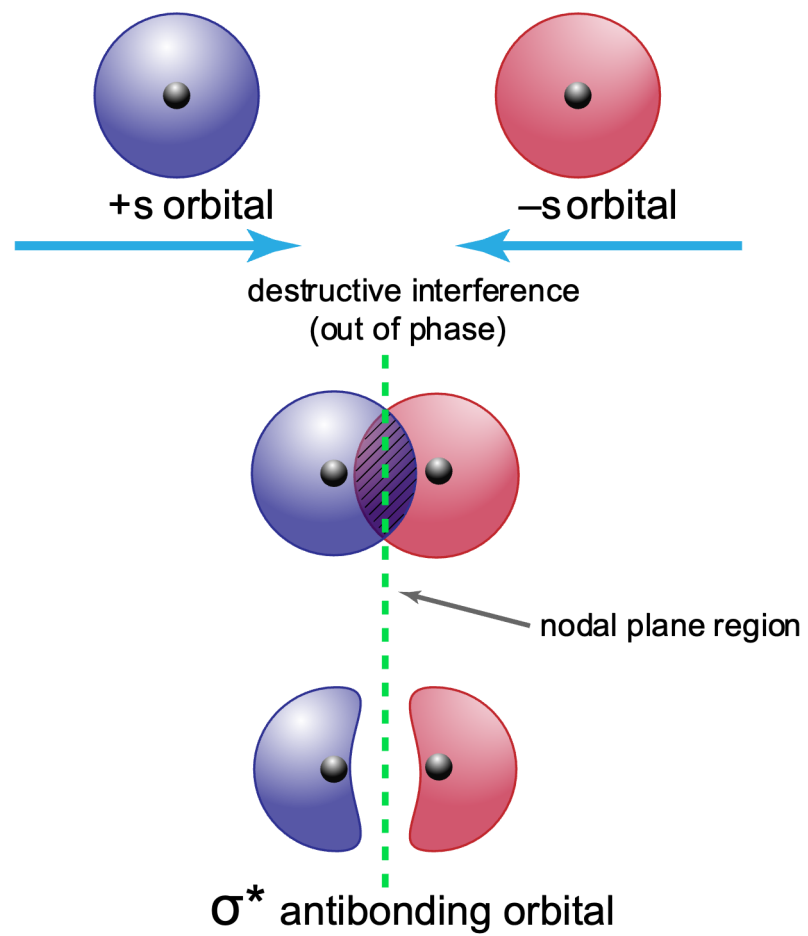


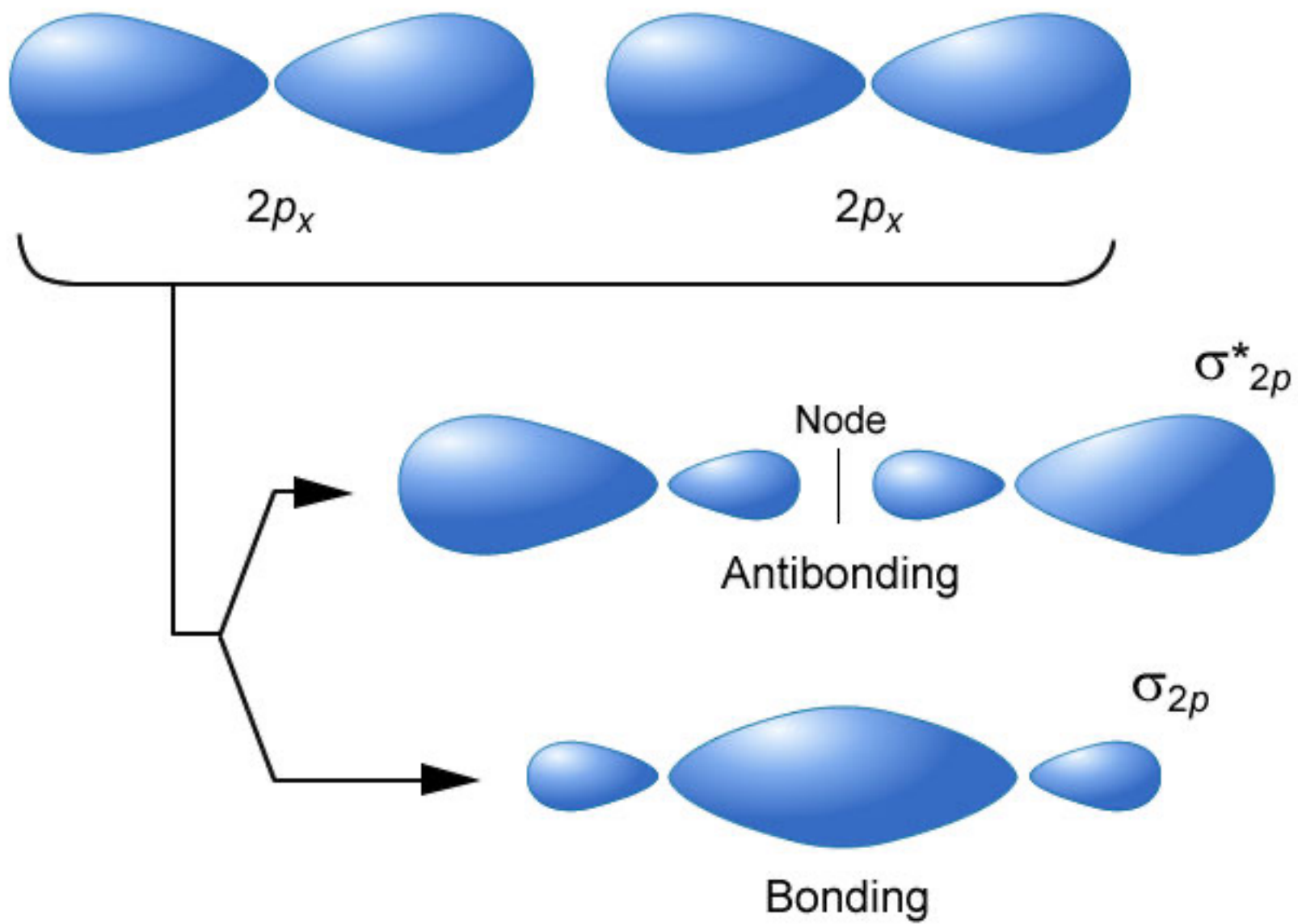
Formation of Molecular Orbitals

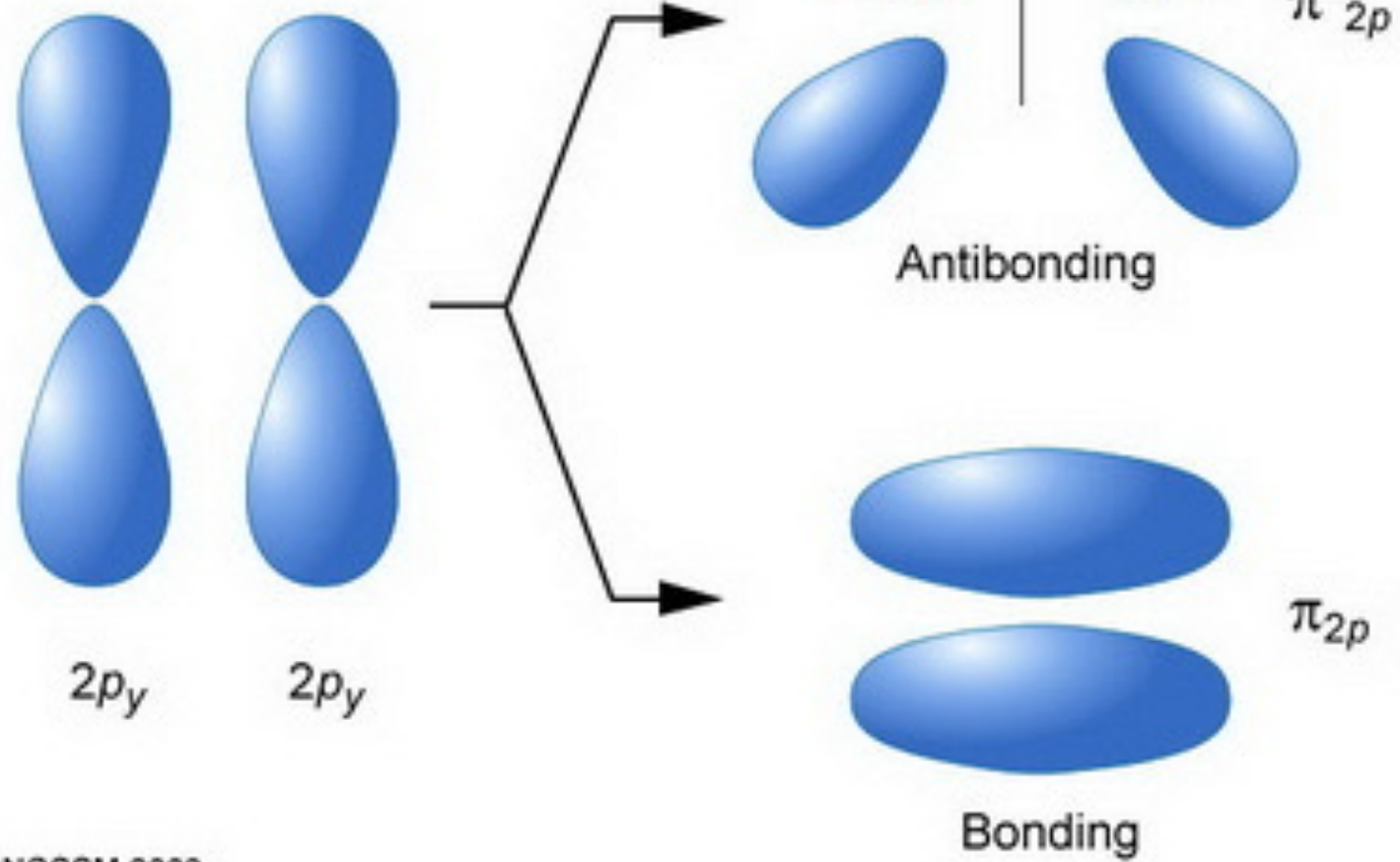
Sigma Bond Formation s - orbitals



sigma *antibond* formation s - orbitals







Molecular Orbitals of O₂ using ORCA and their energies

pKa of an acid and its correlation with physical parameters

Exploring the inductive effect through computational modelling

J. Chem. Educ. 2024, 101, 2815–2822

Learning Objective

- (1) Apply structural and electronic impacts to analyze stability of conjugate bases
- (2) Use computational data to evaluate ranking of acids.

Input File for ORCA

```
# avogadro generated ORCA file  
# HF  
#  
! Opt Freq B3LYP STO-3G CPCM(Water)
```

For molecules, Freq term is needed to get IR spectrum and free energy term

For atoms or ions containing , Freq term is not needed since there is no bond.
Please not down Final single point energy.

Acidity values (pK_a values) are correlated to $\nu(\text{C}=\text{O})$, $\text{C}=\text{O}$ bond lengths, and Mulliken charges for the acidic hydrogen.

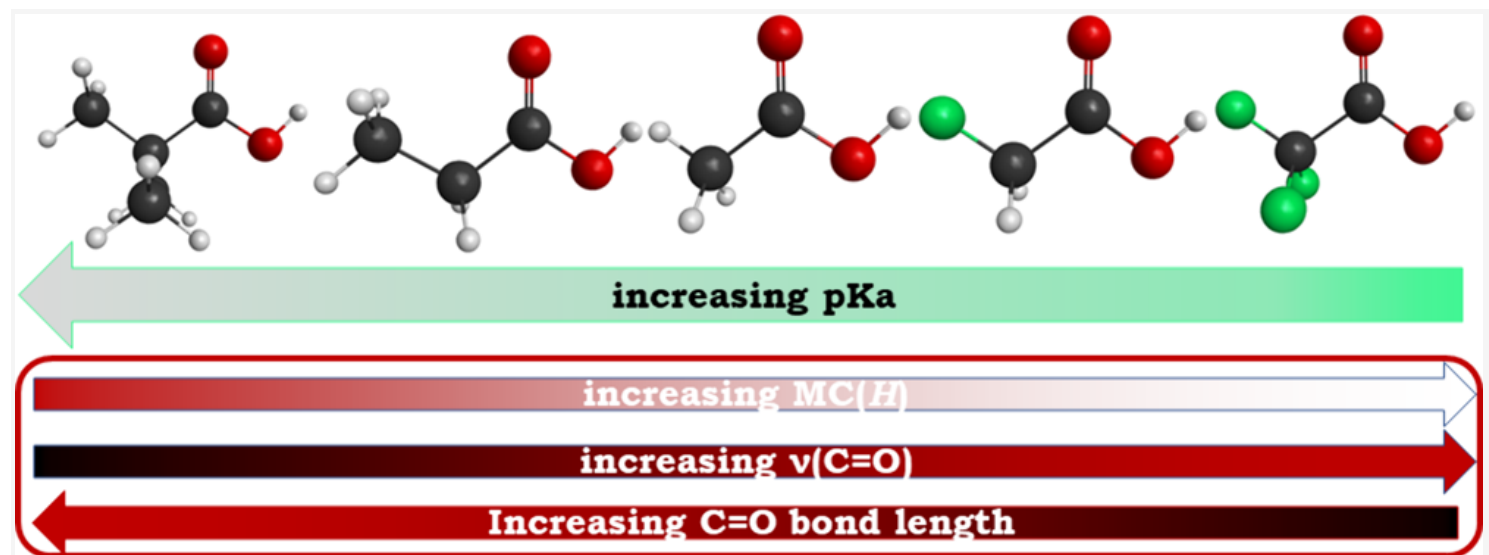
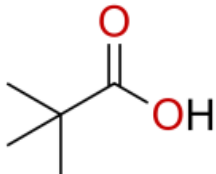
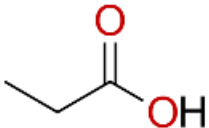
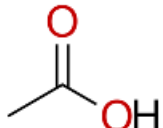
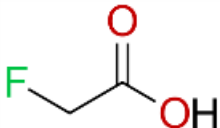
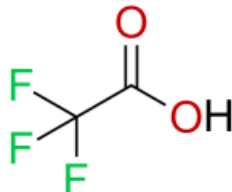
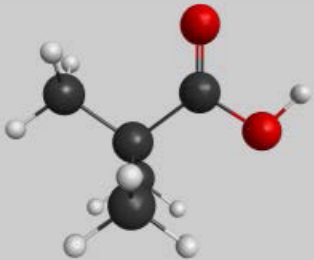
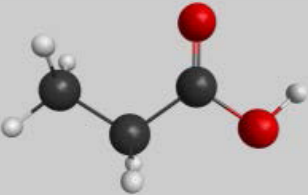
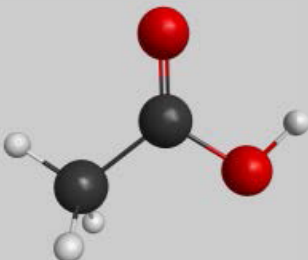
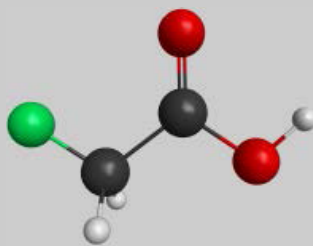
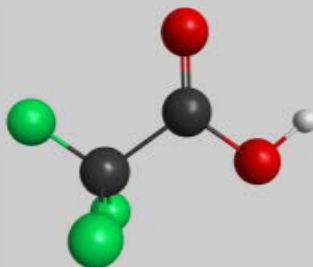


Table 3. Modeled^a Compounds

<i>IUPAC Name</i>	2,2-dimethylpropanoic acid	propanoic acid	ethanoic acid	fluoroethanoic acid	trifluoroethanoic acid
<i>Line Structure</i>					
<i>Optimized Structure*</i>					
pK _a [†]	5.03	4.87	4.76	2.59	0.52
$\nu(\text{C}=\text{O}), \text{cm}^{-1*}$	1767.4	1781.8	1791.9	1813.2	1813.3
C=O, Å [*]	1.2250	1.2236	1.2228	1.2173	1.2158
MC(OH) [*]	0.3939	0.3944	0.3951	0.4038	0.4137

^aAll structures and parameters modeled at the revPBE/6-31G* level of theory. [†]Literature pK_a values obtained from the CRC.²⁶

	HX	H ₂ O	H ₃ O ⁺	X ⁻	dG
HF	-100.47	-76.44	-76.85	-99.98	
HCl	-460.75	-76.44	-76.85	-460.30	
HBr	-2574.55	-76.44	-76.85	-2574.12	
HI	-298.37	-76.44	-76.85	-297.93	

Input file for HF/HCl/HBr/Hi

! Opt Freq revPBE def2-TZVP CPCM(Water)

Input File for F-/Br-/Cl-/I-

! SP revPBE def2-TZVP CPCM(Water)

Calculate ΔG of the reaction

Calculate with different theory like B3LYP in place of revPBE and STO-3G in place of def2-TZVP